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AI-Driven Academic and Career Path Recommender System

Resmika S. K[1], T. Hashni[2]
[1] B.Sc. Artificial Intelligence and Machine Learning,Dr. N.G.P. Arts and Science College, Coimbatore, India
[2] Assistant Professor, Department of Artificial Intelligence and Machine Learning, Dr. N.G.P. Arts and Science College, Coimbatore, India.

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Corresponding Author:
Resmika S. K

Abstract:

The AI-Driven Academic and Career Path Recommendation System is a tool built to help students select an appropriate course of study after 12th grade through a university or college program. The purpose of education is to develop a foundation for future career opportunities by providing individuals with knowledge, skills, and abilities that will shape their future careers. It is often difficult for students to make informed choices about their academic path due to limited information and support, as well as an overwhelming number of possibilities. If students make the wrong choice at this point in their lives, they may experience disappointment and have less opportunity for success in their chosen career. In order to overcome the issue of providing better assistance to students when making educational decisions, this system will use data from machine learning to examine different types of student information to offer recommendations for appropriate courses and colleges. The system will offer students suggested classes and college courses based on what matches their past academic achievement and their preferences. It will exist as a web tool that provides students with a fun and exciting way to view their potential future college/career options.

Keywords: Academic Recommender System - Machine Learning - Random Forest Regressor - College Recommendation - Career Guidance - Artificial Intelligence - Chatbot

1. Introduction

The importance of education cannot be overstated when it comes to developing an individual's knowledge, skills and career. Education forms the basis for an individual's development and success professionally, but also the success and furtherance of society in general. With so many different academic streams, courses and career paths available, the decision-making process regarding what stream of study and/or college to pursue after higher secondary is becoming more challenging for students to determine. Making an informed academic decision can have a tremendous impact on the student's future career opportunities and professional progression.

Many students struggle with making these decisions because they lack awareness of the various options available, they have inadequate guidance to help them make an informed academic choice, and they do not have access to reliable information about colleges/courses to assist them in making their choice. Traditional career counseling methods are typically manual, time-consuming, and generic in nature; these methods do not consider the unique interests, abilities and preferences of each student therefore it is difficult for students to make these decisions. Another factor complicating the ability of students to make sound decisions is that there is no efficient method of comparing colleges based upon easily obtainable factors such as ranking, placement performance, or available courses and/or reputation.

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A proposed AI-Driven Academic Path Recommender System was created to address these issues and serve as an intelligent academic guidance resource for students. The system is a web-based application with a chatbot integrated into it that acts as a virtual academic coach for the student to explore various educational possibilities by providing interactive access to course and college data. The system processes student data (stream of interest; desired area, such as place of residence; preferred course of study; etc.) and assigns recommended colleges based on this data.The system makes use of a Random Forest Regressor model to provide suitability scores to potential colleges. Appropriate suitability scores will be established by the use of factors such as college rankings, placement statistics, and other student-input-dependent behaviour of the institution.

The primary vision of this tool is to provide accurate academic and career recommendations through data-driven analysis and help students choose appropriate streams and colleges by providing information on the available courses, skills necessary for those courses, and potential career paths/occupations after completion of courses taken. Additionally, the system contains a college comparison feature which aids the student(s) in comparing multiple colleges based on how each college is performing using variables such as ranking and placement. Thus enabling the student to effectively identify a college that fits their needs.

2. Literature Review

Several studies have been conducted on the topic of academic and career recommendation systems to assist students in making the right choice of courses and colleges to attend. Machine learning-based recommendation systems have been used to analyze student data and make appropriate recommendations on academics. Machine learning-based recommendation systems have been widely used to analyze student data and make appropriate recommendations on academics. Machine learning helps to make recommendations more accurate; however, the majority of the recommendation systems lack the feature of intelligent ranking to compare the performance of the colleges on the basis of their rankings and placement results.

Career recommendation systems using the rule-based approach have also been developed to provide academic recommendations to the students. The rule-based recommendation system helps to provide recommendations to the students based on their inputs to the system. The rule-based recommendation system is simple and easy to implement; however, the recommendations provided by the system are general and lack personalization. Since the recommendations are based on the rules defined in the system, the system lacks the feature of personalization and is not flexible to accommodate the requirements of the students.

Educational recommender systems using collaborative filtering have also been suggested to recommend courses to the users. These recommender systems suggest courses to the users on the basis of their similarity and the courses chosen in the past. Collaborative filtering is an effective way to recommend courses to the users, provided the user information is sufficient. However, these recommender systems may not be effective in the case of new users who have not chosen courses in the past.

From the analysis of the existing recommender systems, it is observed that the existing recommender systems have limitations such as lack of personalization and the lack of intelligent ranking methods in the existing recommender systems. Moreover, the existing recommender systems rely on historical data to recommend courses to the users. However, the proposed AI-Driven Academic and Career Path Recommender System overcomes the limitations of the existing recommender systems using the machine learning and personalized recommendation methods.

3. Proposed System

The proposed system is a web-based application that aims to assist students in choosing the appropriate academic streams, colleges, and courses using Machine Learning algorithms. It helps the student to explore the various options available to them, compare the colleges, and also provide career guidance through the interactive interface of the proposed system.

3.1 System Architecture

The system has a defined architecture in which the user input is processed through the web interface and the machine learning model to generate the recommendations. First, the user is required to input the information through the Streamlit interface. Then, the input information is processed through the machine learning model and the academic dataset. Finally, the recommendations are generated and the results are displayed to the user.

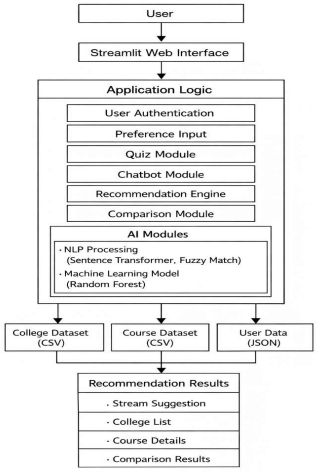


Figure 1: Architecture of the Proposed System

3.2 Module Description

The proposed system consists of different modules, and each module is intended to carry out different functions to provide a comprehensive academic guidance system.

3.2.1 Login Module

The role of the Login Module is to handle user authentication and their respective profiles. It allows new users to create their accounts by registering themselves within the system and allows existing users to log in safely and securely. It saves vital information regarding users such as their name, email address, age, and favorite colleges..

3.2.2 Dashboard Module

The Dashboard Module functions as the interface of the system, which means that it functions as a central hub from which users can access all the other features of the system. After a user has logged into the system, the user is redirected to the dashboard, which means that the user can access all the other modules of the system from the dashboard. It functions as a user-friendly interface through which users can access all the other features of the system, such as the Guider Module, the College Explorer Module, the Recommender Module, and the Comparison Module, among other features of the system, which improves the usability of the system by arranging the features of the system in a proper manner.

3.2.3 Recommender Module

The Recommender Module is the core part of the system that helps create personalized recommendations for colleges. It allows users to choose their academic stream and add preferences such as location, course of interest, and fee structure. Based on these details, colleges are filtered from the database. It employs a Random Forest machine learning algorithm that rates colleges according to significant factors such as their ranking and placement performance. It predicts an AI suitability score for each college and displays them according to their rank based on these predicted scores. This module will help students choose the most suitable colleges according to their requirements.

3.2.4 Quiz Module

The Quiz Module has been designed to assist students in choosing the right academic stream based on their interests and skills. The Quiz Module consists of a set of questions related to academic interests and career aspirations, which students need to answer by choosing the best option according to their interests. Based on the scores, the system suggests the best stream of academic interest to the students. The Quiz Module can be used to suggest the best stream of academic interest, such as Engineering, Medicine, Commerce, Science, etc., to the students. This module would be highly beneficial to students who are confused about their academic interests.

3.2.5 College Explorer Module

This module helps the user search and explore colleges available in the dataset. In addition, it allows the user to apply filters such as stream and location to find suitable colleges. It gives detailed information about each college, enabling students to understand the available options properly. For example, it gives information about the name of the college, city, ranking, fees, performance of placement, and courses offered by the college.

3.2.6 Comparison Module

Comparison Module allows users to compare various colleges simultaneously. In this module, a user can choose two or more colleges and compare their performance according to various significant factors such as ranking and placement. The results of the comparison are represented in a tabular and graphical chart form, making it simpler for users to understand the comparison results. This module will help students compare various colleges and choose the most suitable one according to their performance.

3.2.7 Guider Module

The purpose of the Guider Module is to act as an academic chatbot that gives academic guidance and helps answer questions posed by students. The chatbot helps users pose questions regarding colleges, courses, and career opportunities. The system processes questions posed by the end-user through Natural Language Processing. It makes use of TF-IDF vectorization and cosine similarity to map the query posed by the end-user with existing data in the college and course datasets. It fetches the most relevant data and acts as a response to the query posed by the end-user.

4. Methodology

The methodology of the proposed system includes dataset preparation, data preprocessing, machine learning model development, recommendation generation, and chatbot processing. It helps the system generate accurate and personalized recommendations to the users.

4.1 Dataset

The proposed system utilizes two datasets: the College Dataset and the Course Dataset. Both of these datasets provide the necessary information to the users.

College Dataset

The college dataset includes detailed information about various colleges and institutions. The dataset includes various attributes like the name of the college, the ranking of the college, the placement record of the college, the fee structure of the college, the city of the college, and the streams offered by the college.

Course Dataset

The course dataset includes information about various courses and career options available to the users. The dataset includes attributes like the name of the course, the duration of the course, the skills required to pursue the course, and the job opportunities available after the course.

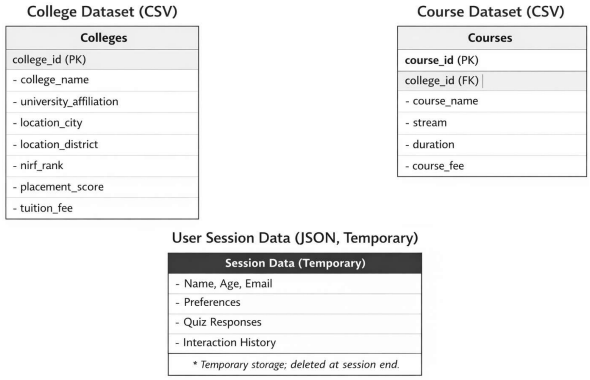


Figure 2: Input Flow of the System

4.2 Data Preprocessing

The data preprocessing is an essential task to ensure that the provided data is accurate and suitable to be fed into the machine learning model. The text values provided in the dataset are processed to remove unwanted characters and spaces. The ranking values provided in the dataset are converted into numerical form to allow the analysis to be done properly using the machine learning model. The placement records provided in the dataset may be in the form of text and are converted into numerical form to calculate the performance of the colleges. The fee values provided in the dataset are converted into average amounts to allow the fee to be compared from one college to another. The missing values provided in the dataset are handled properly by replacing them with the average values or default values.

4.3 Machine Learning Model

The system utilizes a Random Forest Regressor machine learning model for ranking the colleges based on their suitability. Random Forest is a machine learning algorithm that utilizes a combination of decision trees for accurate predictions. The algorithm minimizes errors in predictions. The model utilizes key features such as ranking and placement performance as input variables for predictions. The algorithm generates a score for each college based on the input variables. The score represents the suitability of a college for the student based on the

selected criteria. The system ranks the colleges based on their AI suitability score. The ranked list of colleges is displayed to the user for accurate and reliable results.

The performance of the Random Forest model was assessed through the application of standard regression metrics. It is observed that the model has achieved an *R^2 score of 0.9192,* which reflects the high prediction accuracy of the model. Moreover, the model has achieved a *Mean Absolute Error (MAE) of 0.0234 and Root Mean Square Error (RMSE) of 0.052,* which reflects the low prediction errors of the model and its effective functioning in providing accurate recommendations regarding colleges and their integration into the system.

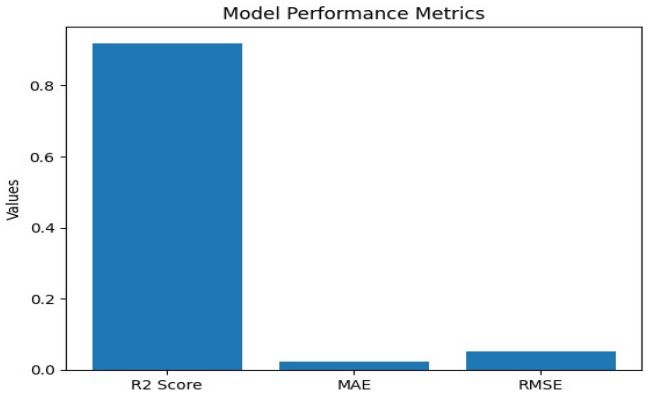


Figure 3: Graphical Representation of Model Evaluation Metrics

4.4 Recommendation Process

In the recommendation process, the user will choose an academic stream and enter their personal preferences such as location and course of interest. Based on this, the system will filter the colleges from the dataset.Once the colleges are filtered, the machine learning model will be used to calculate the AI suitability scores. Then, the colleges will be ranked according to their scores. Finally, the system will display the recommended colleges.

4.5 Chatbot Process

In the chatbot process, academic guidance will be provided by answering questions from the users.In this process, when a user enters a query, it will be processed and converted into numerical vectors using TF-IDF vectorizer.Then, the system will calculate the similarity between the query and content in the dataset using cosine similarity.Based on the similarity scores calculated, the system will choose the query from the dataset and display it to the user.In this process, TF-IDF vectorizer and cosine similarity are used by the chatbot to give accurate and meaningful responses to the users.

5. Implementation

5.1 System Development

The proposed system will be developed as a web application using the Streamlit framework. The application will offer an interactive and user-friendly interface to students to easily access the different features of the system. It will incorporate the application of machine learning techniques in the processing of the datasets to provide students with personalized academic recommendations.The system will have several key components that will enable it to produce accurate results. The User Interface will enable the students to input their details and interact with the different components of the system. The Dataset Processing will handle the processing and storage of the college and course datasets used in the system. The Machine Learning Model will utilize the trained Random

Forest Regressor model to generate AI suitability scores for the colleges. The Recommendation Engine will incorporate the input data, the datasets, and the machine learning model to produce personalized college recommendations.

5.2 System Workflow

The system has a workflow to facilitate the recommendations in an efficient manner. First, the user logs into the system using the login module by providing the required credentials to the system. After the user has successfully logged into the system, the user can access various modules of the system using the dashboard, which acts as a central interface of the system. After accessing the dashboard, the user can choose a module such as the recommender module, college explorer module, comparison module, quiz module, or the guider module, depending upon the requirements of the user.

Once a module is chosen, the system will collect and process the inputs provided by the user such as academic stream and preferences, and search criteria. Then it will collect relevant information from the college and course data set. The collected data will then be processed and filtered according to the requirements of the user. If the recommender module is chosen by the user, the processed data will then be passed through the machine learning model, and AI suitability scores will be calculated and colleges will be ranked accordingly. In the case of the comparison module, analysis of the chosen colleges will be performed and displayed in a suitable manner using tables and charts. For the guider module, analysis of the query provided by the user will be performed using text analysis, and suitable responses will be generated.At last, the processed data will be displayed to the user in an organized manner such as tables and charts. In this way, it will help the user make suitable academic decisions by providing accurate and efficient academic guidance in an easy-to-understand manner.

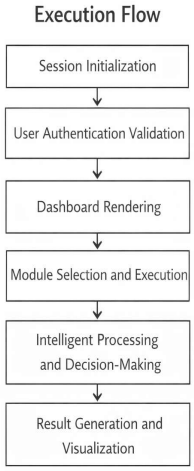


Figure 4: Workflow of the Recommendation System

6. Results

The proposed AI-Driven Academic and Career Path Recommender System has been successfully implemented as a web-based application. The application has been tested using different inputs to check the accuracy and performance of the proposed recommendation process.

The following figures show the working of different modules in the proposed system.

Home Page

The home page is the first interface of the proposed AI-Driven Academic and Career Path Recommender System. The home page introduces the users to the proposed application. The users can get basic information about the proposed application and can proceed to the next page to register themselves.

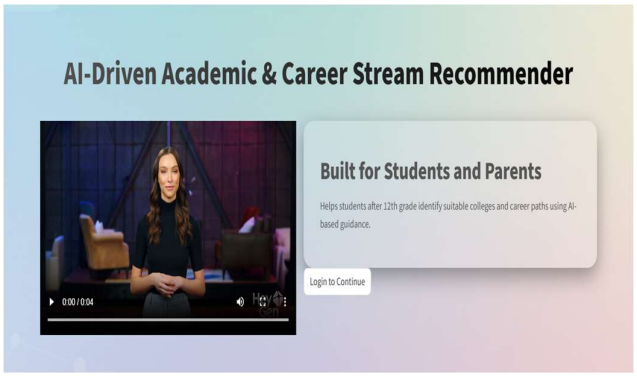


Figure 5: Home Page of the System

Login Page

The proposed Login Page allows users to register and log in to the proposed application. The users can provide the required information to get access to the proposed application.

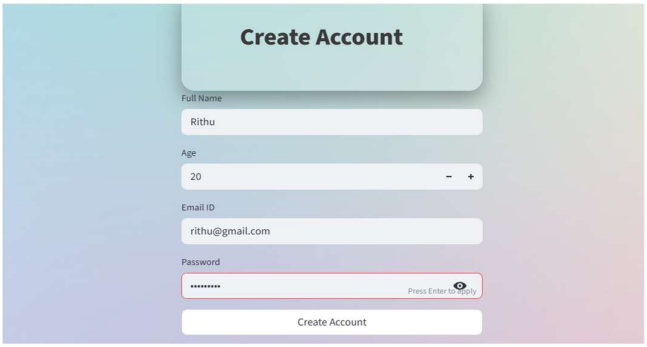


Figure 6: Login Page of the System

Dashboard

The proposed dashboard is the main interface after the user has logged in to the proposed application. The proposed dashboard provides the users with the option to navigate to the different modules such as Recommender, College Explorer, Comparison, and Guider.

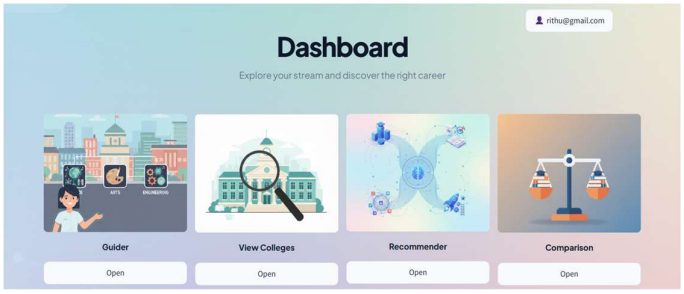


Figure 7: Dashboard Page of the System

Guider Page

The Page is used to guide the user in their academics through an interactive conversation box. Users can ask questions regarding colleges and courses, and the chatbot will give the appropriate information based on the dataset provided

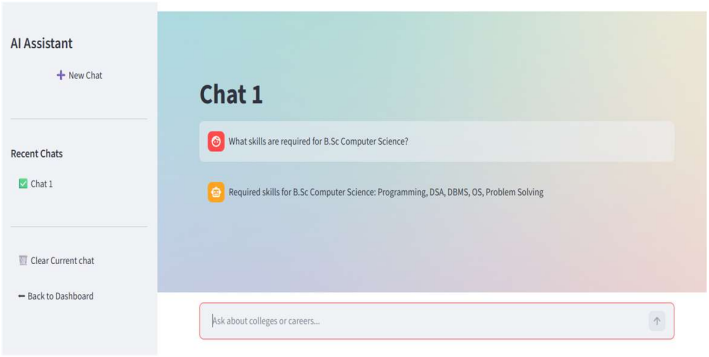


Figure 8: Chatbot Page of the System

Recommender Page

The Recommender Page is used to recommend colleges to the user based on the stream and preferences provided by the user. The system will process the information and display the recommended colleges along with the ranking scores provided by AI.

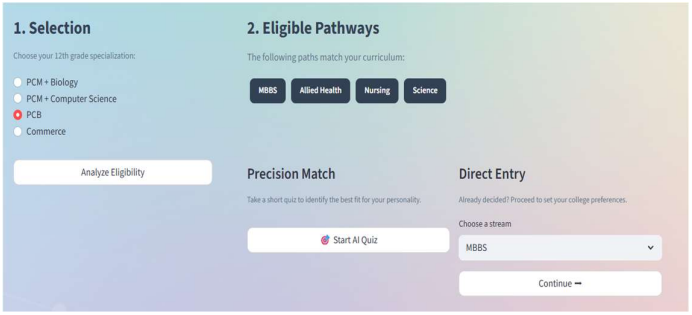


Figure 9: Recommender Page of the System

The results of the quiz will display the recommended stream according to the user's responses to the quiz.

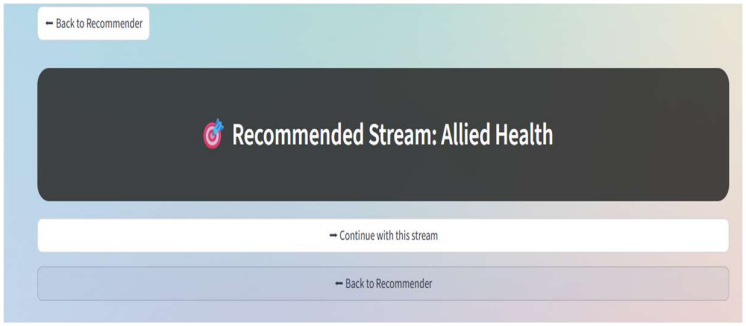


Figure 10: Quiz Module Recommendation for the Streams

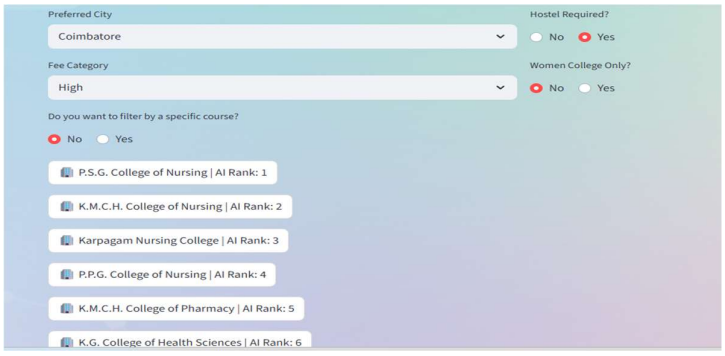


Figure 11: Results of College Recommendation Module

Comparison Page

The Comparison Page is used to compare the colleges provided in the dataset based on the ranking and placement performance of the colleges. The results of the comparison will be displayed in the form of tables and graphical charts.

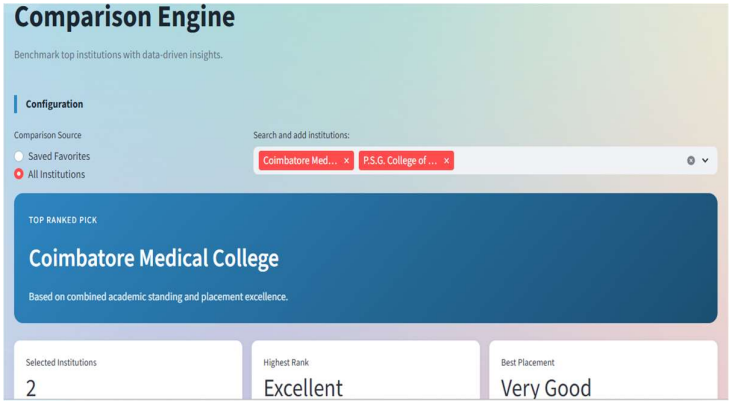


Figure 12: Comparison of the Colleges Module

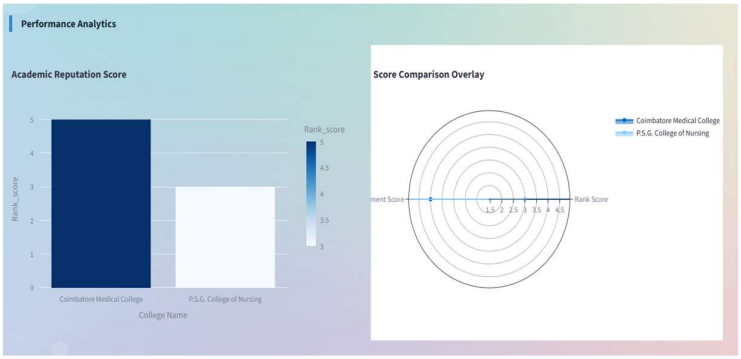


Figure 13 Graphical Comparison of Colleges

7. Conclusion

The AI-Driven Academic and Career Path Recommender System demonstrates the effective use of artificial intelligence in assisting students with academic decision-making. The system provides personalized recommendations for academic streams, courses, and colleges by analyzing student inputs and academic datasets. By integrating machine learning techniques with an interactive web-based interface, the system offers a structured and user-friendly platform for academic guidance. Features such as stream recommendation, college filtering, intelligent ranking, comparison of colleges, and chatbot-based guidance help students explore academic options and make informed decisions.

The system also highlights the importance of combining data-driven recommendations with interactive guidance. The use of a Random Forest model improves the accuracy of college recommendations, while the chatbot module provides additional academic support by answering user queries. Overall, the proposed system simplifies the process of selecting academic paths and demonstrates the potential of AI-based solutions in providing reliable and efficient academic guidance for students.

8. Future Enhancement

The AI-Driven Academic and Career Path Recommender System can also be enhanced further by the addition of other features and the use of advanced technologies. Future developments can include the integration of more data such as international colleges, scholarships, and updated academic data to make the system even better. The AI-Driven Academic and Career Path Recommender System can also be enhanced to include the use of advanced machine learning algorithms to improve the precision of the predictions made by the system.

Future developments can also include the integration of mobile application capabilities to make the system even more accessible to users. The use of multiple languages can also be added to accommodate users with different

linguistic backgrounds. The use of real-time data can also be added to make the data used by the system even more precise and up-to-date. In addition, the integration of user feedback and other career resources such as internships and job opportunities can also make the system even better.

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